

ALLIANCE UNIVERSITY

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NAAC GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Cleaning and Preparing Customer Sales Data Using Power Query

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Cleaning and Preparing Customer Sales Data Using Power Query

Problem Statement

A company has received customer sales data containing missing values, duplicate records, and inconsistent entries. The objective is to clean and prepare the dataset using Power Query before performing business analysis. The dataset used for this experiment is dirty_cafe_sales.csv, containing 10,000 transaction records and 8 fields.

Dataset Overview

Field	Description / Expected Type
Transaction ID	Unique transaction identifier – Text
Item	Cafe product – Text
Quantity	Number of items purchased – Whole Number
Price Per Unit	Unit price – Decimal Number
Total Spent	Transaction value – Decimal Number
Payment Method	Payment type – Text
Location	Sales channel/location – Text
Transaction Date	Transaction date – Date

Methodology / Steps Followed

Step 1: Data Import

Imported the CSV file into Excel Power Query using Data → Get Data → From Text/CSV → Transform Data.

Step 2: Initial Data Inspection

Reviewed the columns and identified missing values, ERROR entries, UNKNOWN entries and data-type issues.

Step 3: Handling Invalid Entries

Replaced invalid ERROR values in numeric/date fields with null and standardized ERROR/UNKNOWN entries in text fields.

Step 4: Handling Missing Values

Handled missing categorical values using a consistent label such as Not Specified/Unknown. Numeric and date values without a reliable replacement were retained as null.

Step 5: Text Cleaning

Applied Trim and Clean transformations to remove unnecessary spaces and unwanted characters.

Step 6: Data Type Conversion

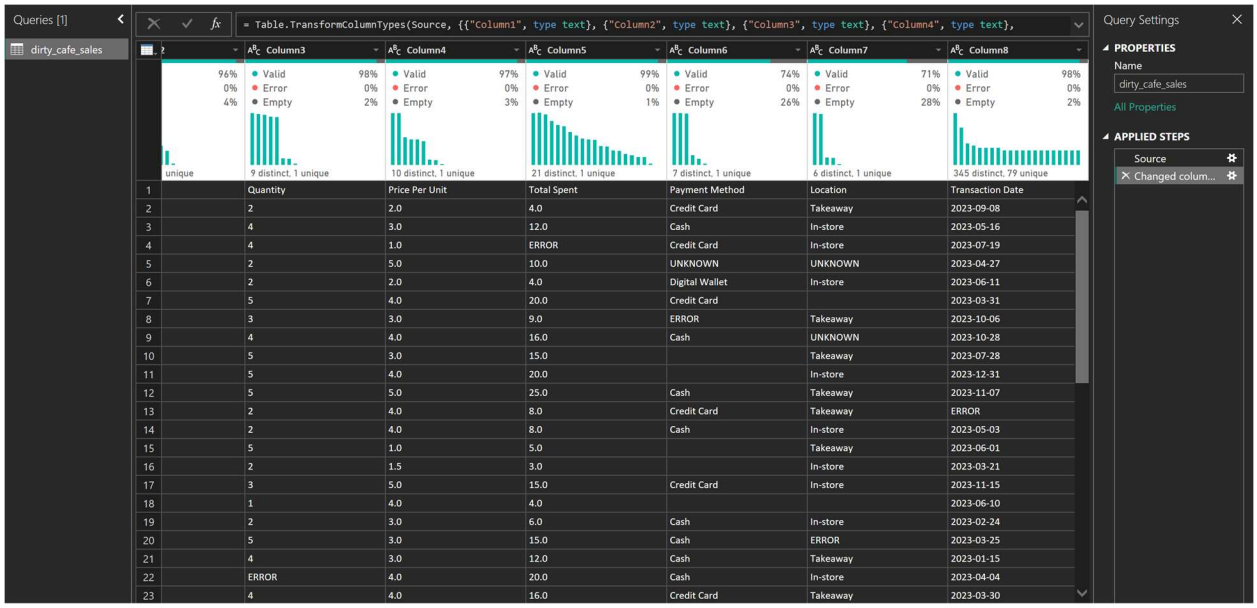
Assigned Text, Whole Number, Decimal Number and Date types to the appropriate columns after cleaning invalid entries.

Step 7: Duplicate Check

Used Remove Duplicates with Transaction ID to check for repeated transaction records.

Step 8: Final Validation

Verified that the cleaned preview contained 0% errors in the visible cleaned fields and was suitable for analysis.



	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
2	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
3	4	3.0	12.0	Cash	In-store	2023-05-16
4	4	1.0	ERROR	Credit Card	In-store	2023-07-19
5	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
6	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
7	5	4.0	20.0	Credit Card		2023-03-31
8	3	3.0	9.0	ERROR	Takeaway	2023-10-06
9	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
10	5	3.0	15.0		Takeaway	2023-07-28
11	5	4.0	20.0		In-store	2023-12-31
12	5	5.0	25.0	Cash	Takeaway	2023-11-07
13	2	4.0	8.0	Credit Card	Takeaway	ERROR
14	2	4.0	8.0	Cash	In-store	2023-05-03
15	5	1.0	5.0		Takeaway	2023-06-01
16	2	1.5	3.0		In-store	2023-03-21
17	3	5.0	15.0	Credit Card	In-store	2023-11-15
18	1	4.0	4.0			2023-06-10
19	2	3.0	6.0	Cash	In-store	2023-02-24
20	5	3.0	15.0	Cash	ERROR	2023-03-25
21	4	3.0	12.0	Cash	Takeaway	2023-01-15
22	ERROR	4.0	20.0	Cash	In-store	2023-04-04
23	4	4.0	16.0	Credit Card	Takeaway	2023-03-30

Fig 1:- Raw Cafe Sales Dataset Before Cleaning

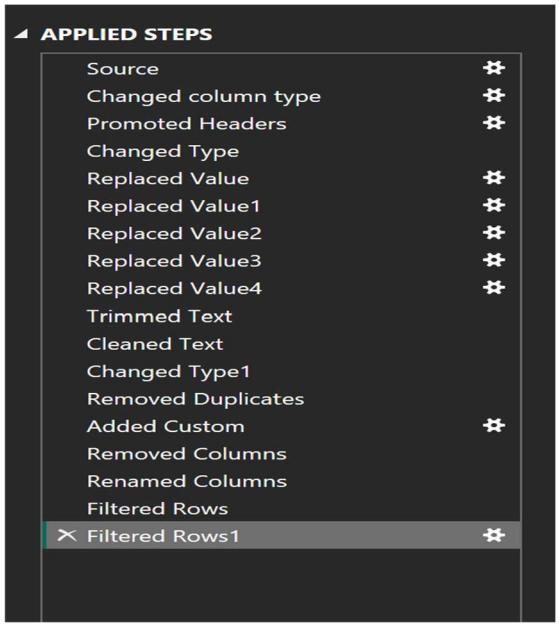


Fig 2: Data Cleaning and Transformation Steps in Power Query

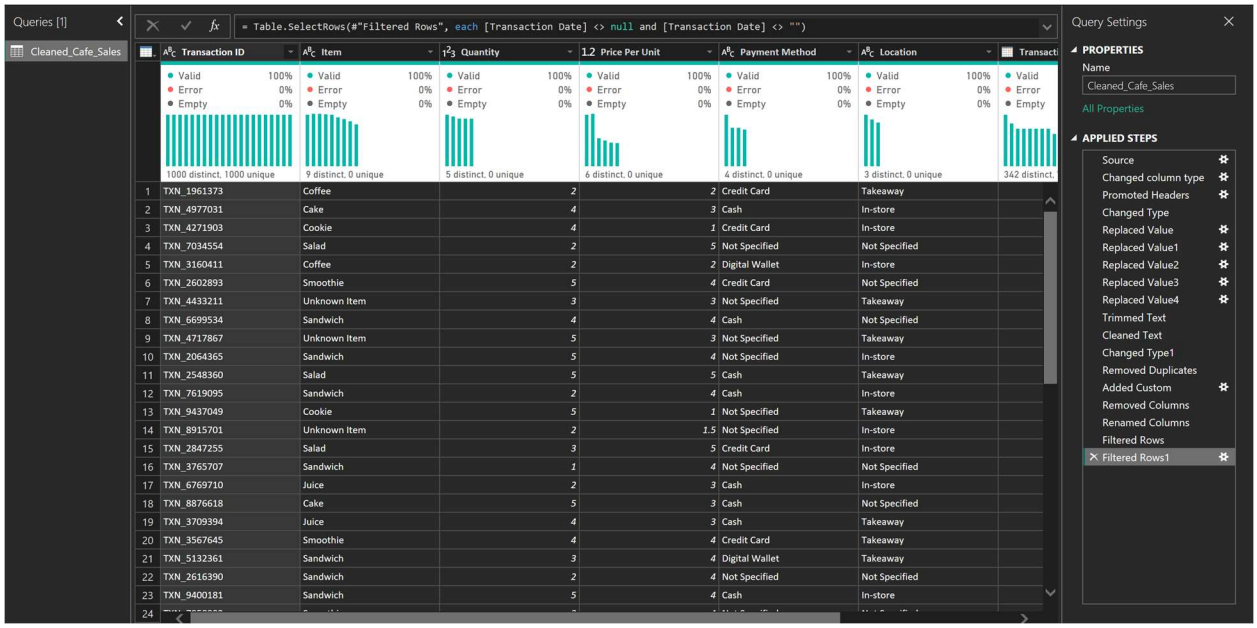


Figure 3: Final Cleaned Dataset and Data Quality Check

Output

The Power Query transformation produced a cleaned dataset suitable for further analysis. The final preview showed no data-type errors in the cleaned fields. Some numerical and date fields retained null values where the original information was missing and could not be reliably inferred.

Cleaning Summary

Issue	Action Taken	Result
ERROR values	Replaced invalid numeric/date entries with null and standardized text errors.	No visible errors
UNKNOWN values	Standardized to a consistent missing-value label.	Consistent categories
Blank categorical values	Handled as Not Specified/Unknown.	No empty categorical fields
Missing numeric/date values	Retained as null when no reliable replacement was available.	No false values
Text inconsistencies	Trimmed and cleaned text.	Standardized text
Duplicate records	Checked/removed using Transaction ID.	Duplicate check completed
Data types	Converted columns to appropriate types.	Analysis-ready schema

Power Query Cleaning Workflow

Raw CSV	→	Inspect	→	Clean ERROR / UNKNOWN
↓		↓		↓
Missing Values	→	Trim & Clean	→	Correct Data Types
↓		↓		↓
Remove Duplicates	→	Validate	→	Clean Dataset

Final Data Quality Check

The final Power Query preview showed 0% errors for Quantity, Price Per Unit, Total Spent, Payment Method, Location and Transaction Date. Quantity, Price Per Unit and Transaction Date still contained some empty values because the source data had missing information. These were preserved as null where a trustworthy replacement was unavailable.

Sample Cleaned Records

Quantity	Price/Unit	Total Spent	Payment Method	Location	Date
2	2.0	4	Credit Card	Takeaway	08-09-2023
4	3.0	12	Cash	In-store	16-05-2023
4	1.0	4	Credit Card	In-store	19-07-2023
2	5.0	10	Not Specified	Not Specified	27-04-2023

Conclusion

The cafe sales dataset was successfully cleaned and prepared using Power Query. The process addressed major data-quality issues including missing values, ERROR entries, UNKNOWN entries, text inconsistencies, data-type problems and duplicate records. The cleaned table is now structured for reliable business analysis and reporting.

Learning Outcomes

Through this experiment, I learned:

- How to import a CSV dataset into Power Query.
- How to identify and handle missing and invalid values.
- How to standardize inconsistent categorical entries.

- How to trim and clean text fields.
- How to assign appropriate data types to columns.
- How to check and remove duplicate transaction records.
- How Power Query prepares raw business data for analysis.
- How to validate a cleaned dataset before reporting.

Power Query Operations Used

Operation	Purpose
Replace Values	Standardize ERROR and UNKNOWN entries
Trim / Clean	Remove unwanted spaces and characters
Change Type	Assign correct data types
Remove Duplicates	Prevent repeated transaction records
Custom Transformation	Support data validation/calculation where appropriate
Close & Load	Load the prepared dataset into Excel

Result

The final cleaned dataset can now be used for analysis such as item-wise sales, payment-method analysis, location/channel analysis, transaction trends and other business reporting activities.